

Wine Questionnaire Rule Engine Handbook (Expanded)

1. Rule Structure

Every rule has:

source = triggering question_key

target = affected question_key

action = what to do

Optional keys depend on the action (mapping, when, options, value...).

2. FILTER Action

Purpose: Restrict available options.

Example

source = country

target = sub_region

action = filter

France -> [Loire Valley, Bordeaux, Burgundy]

Italy -> [Tuscany, Veneto]

Flow:

User selects France

→ Engine loads mapping['France']

→ Allowed options become Loire Valley, Bordeaux, Burgundy

→ All remaining options are removed.

3. HIDE_OPTION Action

Purpose: Hide only specific options.

Example

source = wine_type

target = sub_region

action = hide_option

when = Red

options = [Champagne]

Flow:

User selects Red

→ Engine keeps all filtered options

→ Champagne is removed

→ Remaining French regions stay visible.

4. SKIP_QUESTION Action

Purpose: Skip an entire question.

Example

```
source = wine_type  
target = tannin  
action = skip_question  
when = White
```

Flow:

User selects White

→ shouldSkipQuestion() returns true

→ Tannin question is never rendered.

5. Multiple Rules Together

Example:

Rule 1:

Country=France

→ Filter sub-regions

Rule 2:

Wine Type=Red

→ Hide Champagne

Result:

Loire Valley

Bordeaux

Burgundy

Rhone Valley

...

Champagne removed.

The engine executes both rules automatically.

6. Adding A New Rule

Client Request:

'If Country = India, show only Nashik and Nandi Hills.'

Solution:

Use existing FILTER action.

No JavaScript changes required.

Only questionnaire_rules.php is updated.

7. Adding A New Action

Client Request:

'If Sparkling is selected, automatically choose Celebration.'

Existing actions cannot do this.

Create:

action = set_default

Example:

source = wine_type
target = occasion
action = set_default
when = Sparkling
value = Celebration

Then add a new case:

```
case 'set_default':  
this.applySetDefault(...);
```

No existing rules need modification.

8. Future Action Examples

show_question
Show Food Pairing only for Red wines.

clear_answer
Country changes → clear previously selected Sub Region.

require
If Budget > 5000 → Region becomes mandatory.

disable_option
Show Champagne but disable selection.

sort_options
Display India first for Domestic region.

message
Show:
'Great choice! Red wines pair well with steak.'

9. Debugging

If a rule fails:

- ✓ Check question_key
- ✓ Check template id
- ✓ Check source answer
- ✓ Check database option text
- ✓ Check questionnaire_rules.php text
- ✓ Ensure strings match exactly
- ✓ Confirm action exists in QuestionnaireEngine

10. Best Practices

- Never use question text.
- Always use question_key.

- Rules belong in questionnaire_rules.php.
- Business logic should not be hardcoded in JavaScript.
- Database values and rule values must match exactly.
- Add new functionality by creating new actions instead of modifying old ones.